

PHASE 2

Data Validation & Data Modeling

2016 – 2025 NFL Seasons

Following the completion of the automated data pipeline in Phase 1, Phase 2 focused on transforming the collected datasets into a production-ready analytical data model. Rather than immediately importing data into SQL or Power BI, every dataset was validated, standardized, cleaned, and modeled to ensure consistency, integrity, and long-term maintainability. The objective was to replicate the data quality and modeling practices used by business intelligence teams before reporting and visualization begin.

Step 1 — Dataset Validation

The first step was a comprehensive validation audit across every master dataset generated during Phase 1. Rather than manually inspecting spreadsheets, reusable Python validation scripts produced detailed reports documenting every issue discovered across all five datasets.

- Chicago Bears Team Statistics
- Chicago Bears Game Logs
- Chicago Bears Player Statistics
- Chicago Bears Salary & Contract Data
- NFL League Statistics

Each dataset was evaluated against the following validation checks:

- | | |
|--------------------------|---------------------------------|
| ✓ Total row count | ✓ Season validation (2016–2025) |
| ✓ Total column count | ✓ Data type verification |
| ✓ Duplicate rows | ✓ Team abbreviation consistency |
| ✓ Duplicate primary keys | ✓ Player name standardization |
| ✓ Missing values | |

Step 2 — Data Cleaning

Validation identified issues requiring correction before the data could be used for analytics. The salary dataset and player statistics dataset each required targeted cleaning.

Salary Dataset

The salary dataset initially contained duplicate contract records resulting from historical contract revisions and repeated player entries. Cleaning removed exact duplicate rows and duplicate player-season records,

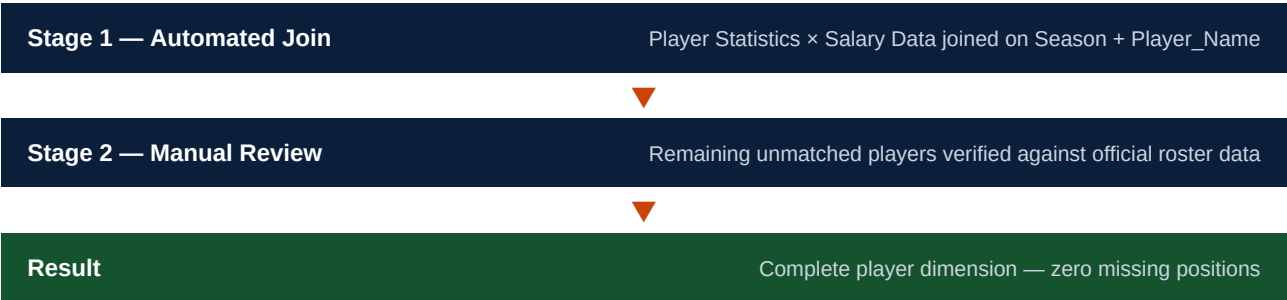
preserving the most complete contract information and maintaining one unique record per Season + Player_Name combination.

Metric	Before	After
Duplicate Rows	48	0
Duplicate Primary Keys	280	0

The resulting dataset satisfies a true primary key structure suitable for SQL and Power BI relationships.

Player Position Enrichment

The player statistics dataset originally lacked position data because the play-by-play source did not include player positions. A two-stage enrichment process was implemented to resolve this.



Stage 2 unmatched records were typically caused by players added after the historical contract snapshot, abbreviated names, recent roster additions, or name formatting differences. These were verified against official roster information and manually assigned where necessary, producing a complete player dimension with no missing positions.

Step 3 — Data Standardization

Every dataset was standardized to ensure consistency across the analytical model.

Seasons

All records verified to fall within the 2016–2025 window.

Team Abbreviations

Every Chicago Bears record standardized to CHI. League-wide abbreviations validated across all 32 franchises.

Player Names

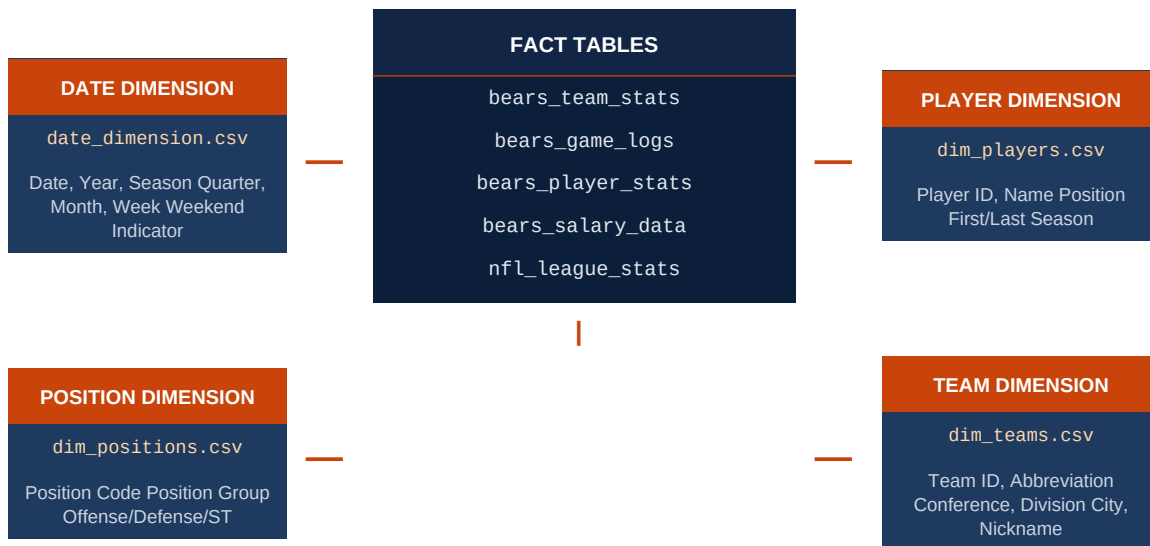
Leading/trailing whitespace removed, duplicate spaces eliminated, and naming conventions normalized to improve matching across datasets.

Data Types

Numeric fields — including wins, losses, passing yards, contract value, and guaranteed money — verified for correct integer or decimal formatting to prevent import issues in SQL and Power BI.

Step 4 — Dimensional Data Modeling

With validated fact tables in place, a star schema dimensional model was created to support efficient analytical querying and Power BI relationships. Four dimension tables were generated alongside the five existing fact tables.



Date Dimension

A complete calendar table covering every day from January 1, 2016 through December 31, 2025. Includes date, year, season, quarter, month, week, day of week, and weekend indicator.

Player Dimension

One record per player containing Player ID, player name, position, first season, and last season. Player IDs provide a stable key for cross-dataset relationships.

Position Dimension

Unique football positions organized into business-friendly groups: Offense (QB, RB, WR, TE), Defense (CB, LB, DL, S), and Special Teams (K, P, LS). Supports filtering and aggregated reporting by position group.

Team Dimension

A complete NFL team lookup table for all 32 franchises including Team ID, abbreviation, full name, conference, division, city, and nickname. Provides a reusable reference for league-wide comparisons.

Step 5 — Dimension Validation

After generating the dimension tables, a second validation pass confirmed the analytical model remained internally consistent. Checks included unique primary keys, no duplicate dimension members, no missing relationships, valid position groups, and complete team and player coverage. Every table passed.

Phase 2 Deliverables

At the completion of Phase 2, the project contains a fully validated analytical data model consisting of five fact tables and four dimension tables. Fact rows are shaded white; dimension rows are shaded orange.

Dataset	File	Type
Team Statistics	bears_team_stats_2016_2025.csv	Fact
Game Logs	bears_game_logs_2016_2025.csv	Fact
Player Statistics	bears_player_stats_2016_2025.csv	Fact
Salary Data	bears_salary_data_2016_2025.csv	Fact
League Statistics	nfl_league_stats_2016_2025.csv	Fact
Date Dimension	date_dimension.csv	Dim
Player Dimension	dim_players.csv	Dim
Position Dimension	dim_positions.csv	Dim
Team Dimension	dim_teams.csv	Dim

Validation Results

Final validation across all nine tables produced zero errors and zero warnings.

Table	Dataset	Result
Fact	Team Statistics	PASS
Fact	Game Logs	PASS
Fact	Player Statistics	PASS
Fact	Salary Data	PASS
Fact	League Statistics	PASS
Dimension	Date	PASS
Dimension	Players	PASS
Dimension	Positions	PASS
Dimension	Teams	PASS

9	0 Tables Validated	0 Total Errors	100% Total Warnings
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Outcome

Phase 2 transforms Bearlytics from a collection of clean datasets into a structured analytical data model ready for enterprise reporting. Through automated validation, data cleansing, dimensional modeling, and relationship preparation, the project now mirrors the foundational stages of a professional business intelligence solution.

Rather than importing raw CSV files directly into reporting software, the project follows a disciplined workflow in which data quality is verified before analysis begins. This ensures that future SQL queries, Power BI relationships, DAX measures, and executive dashboards are built on consistent, validated, and reproducible data.

Project Roadmap



Green = complete. Orange = current phase. Remaining phases cover SQL analysis, Power BI DAX modeling, executive dashboard development, and GitHub portfolio presentation.